

Technical Report: Capacitor Inductance Measurement Based on Low Voltage Surge Discharge Method

Document Metadata

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Report / Project Nr.	69	Edition	1
Company	TDK / EPCOS Electronic Components, S.A.	Status	Confidential / Proprietary

1. Introduction

Capacitor inductance measurement is in general complex due to the fact that the parasitic inductance value (L_s) is extremely low, typically in the nanohenry (nH) range. It is possible to measure this parasitic inductance using a component analyzer; however, it is mandatory to properly compensate the test wires by performing a short/open test prior to measurement. Furthermore, the complete system (capacitor + wires) must remain entirely undisturbed and unmoved during the duration of the test.

When low capacitor inductance values must be measured, or when the capacitor possesses more than one pair of terminals (requiring an external busbar for the measurement setup), it is strongly recommended to use the **Low Voltage Surge Discharge Method**, at least to confirm the accuracy of the values obtained with the component analyzer.

2. Theoretical Framework

The low voltage surge discharge inductance measurement method is based on the physical principle that a capacitor always discharges at the natural resonance frequency of the circuit loop. To isolate the capacitor's internal properties, the discharge must be provoked by short-circuiting its terminals with the minimum possible external loop impedance (both inductance and resistance).

The inductance of the external short-circuiting element is subsequently subtracted from the total loop inductance derived from the captured voltage waveform. Terminal geometry dictate specific preparation criteria:

- Two-terminal Configurations:** In case the capacitor has only two terminals, it can be discharged directly with a copper plate that is as short as possible and possesses a generous cross-section (e.g., MKP capacitors).
- Multi-terminal Polarity Configuration:** If the capacitor has more than one terminal per polarity, and terminals

with the same polarity are connected via an external busbar in the actual application, it is necessary to use an overlapped external busbar to connect them. This configuration ensures that the loop inductance from the precise point where the short-circuit is provoked to all pairs of terminals remains equal (e.g., most MKK capacitors).

- **Integrated Busbar Designs:** In case the capacitor features an integrated busbar designed to connect directly to semiconductor modules, the inductance must be evaluated independently at each pair of terminals (e.g., PCC-LP capacitors).

2.1 Circuit Modeling and Mathematical Derivations

Following Kirchhoff's voltage law and solving the linear differential equation of the resulting RLC circuit under low-damping conditions, the following expressions for the transient response of current and voltage are established:

Equation 1: Damping Factor Ratio via Consecutive Peaks

Dividing the current equations at two consecutive maxima yields the logarithmic decrement relationship:

$$e^{(a * T)} = I_{p1} / I_{p2}$$

Where I_{p1} is the first peak current, I_{p2} is the second consecutive peak current, and T is the period of oscillation.

Equation 2: Damping Coefficient Expression

The damping coefficient (a) is defined as:

$$a = R / (2 * L)$$

Where R is the total loop resistance ($R = R_d + R_c$) and L is the total loop inductance ($L = L_d + L_c$). Here, subscript d refers to the discharger switch/plate and subscript c refers to the internal capacitor parasitics.

Equation 3: Natural Angular Frequency Relation

The undamped natural frequency (ω_0) relates to the damped resonance frequency (ω_r) and damping coefficient (a) by:

$$\omega_0 = \sqrt{(\omega_r^2 + a^2)}$$

Equation 4: Ideal LC Frequency Relationship

The relationship between undamped angular frequency and total circuit components is given by:

$$\omega_0 = 1 / \sqrt{L * C}$$

Equation 5: Internal Capacitor Inductance Derivation

The ratio of the initial inductive voltage drop (V_1) to the initial charging voltage (V_c) allows the segregation of capacitor inductance (L_c) from the total loop inductance (L):

$$V_1 / V_c = L_c / L$$

3. Experimental Procedure & Guidelines

Although the initial charging voltages utilized for this procedure are very low ($< 20V$), the low total circuit loop inductance causes the peak discharge currents to reach high magnitudes, particularly when evaluating

high-capacitance, low-inductance components (peaks can reach up to several kA).

As a preliminary estimation, this peak discharge current can be approximated using the following idealized formula:

$$I_{p1} = V_c * \sqrt{C / L}$$

3.1 Operational Constraints

- **Voltage Constraints:** As a general operating rule, V_c must be maintained as low as possible to acquire crisp, clean current and voltage waveforms. Recommended charging voltage values should reside above 4V and strictly below 20V.
- **Sensor Selection:** It is crucial to employ appropriate high-bandwidth current and voltage sensors whose amplitude ranges are scaled to match expected experimental limits. A **Rogowski coil** must be used for monitoring current waveforms.
- **Data Acquisition Setup:** Both instrumentation sensors must be connected simultaneously to the same digital oscilloscope. The transient capture should be triggered using the voltage drop as the primary triggering factor.
- **Time Scale Setting:** The oscilloscope time base must be configured according to the expected waveform period T , estimated from the idealized equation below (the actual period will be slightly longer due to physical damping parameters):

$$T = 2 * \pi * \sqrt{L * C}$$

4. Practical Measurement Example

To demonstrate the execution of this protocol, an experimental test was carried out on an AC film capacitor with the following specifications:

- **Component Details:** EPCOS Metallized Polypropylene (MKP) Film Capacitor
- **Nominal Ratings:** 650 μ F, 350 VDC
- **Temperature Range:** -25°C to +65°C
- **Serial Identification:** B323201 / 110459773

4.1 Raw Experimental Parameters Extracted from Waveforms

The discharge waveforms acquired from the oscilloscope yielded the following parameter values:

- Peak Current 1 (I_{p1}): 92 A
- Peak Current 2 (I_{p2}): 22.5 A
- Oscillation Period (T): 16.2 μ s
- Initial Charging Voltage (V_c): 5.04 V
- Inductive Drop Voltage Component (V_1): 2.72 V
- Ohmic/Damped Voltage Component (V_2): 2.32 V
- Measured Real Capacitance (C): 63.3 μ F

4.2 Step-by-Step Inductance Calculation

Step 1: Calculate the Damping Coefficient (a)

Applying Equation 1:

$$e^{(a * 16.2\mu s)} = 92 / 22.5 \Rightarrow a = 87,097 \text{ rad/s}$$

Therefore, the damping term yields the circuit balance relationship: $R / (2L) = 87,097\text{ }\Omega$.

Step 2: Determine Resonance and undamped Natural Frequencies

The damped angular resonance frequency (ω_r) is computed as:

$$\omega_r = (2 * \pi) / T = (2 * \pi) / 16.2\mu s = 387,864 \text{ rad/s} \Rightarrow f_r = 61,730 \text{ Hz}$$

The undamped natural frequency (ω_0) is derived via Equation 3:

$$\omega_0 = \sqrt{(387,864^2 + 87,097^2)} = 397,527 \text{ rad/s}$$

Note: Because high-quality film capacitors exhibit inherently low equivalent series resistance (ESR), the damping coefficient remains small, meaning the resonance and natural frequencies are closely aligned.

Step 3: Total Loop Inductance Calculation (L)

Using Equation 4 and the measured capacitance ($C = 63.3\text{ }\mu\text{F}$):

$$\omega_0 = 1 / \sqrt{(L * C)} \Rightarrow 397,527 = 1 / \sqrt{(L * 63.3\mu F)} \Rightarrow L = 100.1 \text{ nH}$$

Step 4: Segregation of Internal Capacitor Inductance (L_c)

Applying Equation 5 with the measured voltage criteria:

$$V_1 / V_c = L_c / L \Rightarrow 2.72V / 5.04V = L_c / 100.1 \text{ nH} \Rightarrow L_c = 54 \text{ nH}$$

The final calculated internal parasitic inductance of the capacitor under test is **54 nH**.

5. Simulation Verification and Validation

A primary benefit of the low voltage surge discharge technique is that after testing, results can be verified using circuit simulation software. By inputting the calculated parameters back into a standard RLC network model, simulated waveforms can be directly compared with real oscilloscope captures to confirm the validity of the results.

The total extracted loop resistance from the mathematical solution is $R = 17.4\text{ m}\Omega$. Given that the internal Equivalent Series Resistance (R_c , or ESR) at the discharge frequency ($61,730\text{ Hz}$) is known to be $9.78\text{ m}\Omega$, the external discharger loop resistance (R_d) is isolated by subtraction:

$$R_d = R - R_c = 17.4 \text{ m}\Omega - 9.78 \text{ m}\Omega = 7.62 \text{ m}\Omega$$

Re-running the network simulation with these synthesized constraints ($L = 100.1\text{ nH}$, $C = 63.3\text{ }\mu\text{F}$, and $R = 17.4\text{ m}\Omega$) yields simulated waveforms that closely match experimental captures.

5.1 Consolidated Results Table

Parameter Description	Simulated Value	Tested / Real Value	Deviation Summary
First Peak Current (I_p1)	93 A	92 A	+1.08%

Parameter Description	Simulated Value	Tested / Real Value	Deviation Summary
Second Peak Current (I_p2)	22.8 A	22.5 A	+1.33%
Oscillation Period (T)	16.2 μs	16.2 μs	0.00% (Identical)
Inductive Voltage Component (V_1)	2.71 V	2.72 V	-0.37%
Damped Voltage Component (V_2)	2.33 V	2.32 V	+0.43%

Because the simulated waveforms match the real oscilloscope measurements with high precision, the calculation methodology and the isolated internal capacitor parasitic inductance value of **54 nH** are verified as correct.